KIGALI INDEPENDENT UNIVERISTY



GROUP QUIZ OF DATABASE MANAGEMENT SYSTEM

By Mrs Odette.

STUDENT ROLLNUMBERS:

- ✓ **202212040**
- ✓ **202211528**
- ✓ **202211627**
- ✓ **202211258**
- ✓ **202211373**
- ✓ **202211054**
- ✓ **202211745**
- ✓ **202211215**

a) Normalize this table up to the 3rd Normal Form (3NF).

To normalize this table to 3NF:

Step 1: Identify functional dependencies

- Proj_Id → Proj_Name
- Emp_Name → Position, Payment_Per_Hour
- Emp_Name, Proj_Id → Tot_Hours, Tot_Payment

Step 2: Break down the table

1. **Projects Table:**
    - Proj_Id (Primary Key)
    - Proj_Name
2. **Employees Table:**
    - Emp_Id (Primary Key) *(add unique ID for employees)*
    - Emp_Name
    - Position
    - Payment_Per_Hour
3. **Work Table:**
    - Work_Id (Primary Key) *(add unique ID for work records)*
    - Emp_Id (Foreign Key referencing Employees.Emp_Id)
    - Proj_Id (Foreign Key referencing Projects.Proj_Id)
    - Tot_Hours
    - Tot_Payment

1. Projects Table

Proj_id	Proj_name
01	Harvest
05	ToT

2. Employees Table

Emp_id	Emp_Name	Position	Payment_per_Hour
101	Bob	IT	$30
102	Tom	Engineer	$15
103	Peter	Engineer	$15
104	Diana	Engineer	$15

3. **Employees project table**

Emp_id	Proj_id	Tot_Hours	Tot_payment
101	01	150	$3600
102	01	210	$3900
103	01	100	$1900
104	05	210	$2800

b. Let's display the names of employees who earn a total payment between 3,500 and 8,000.

First let's use the sql query then we can show how that table looks like:

SELECT e.Emp_Name, w.Tot_Payment **FROM** Employees e JOIN Work w ON e.Emp_Id = w.Emp_Id

WHERE w.Tot_Payment **BETWEEN** 3500 AND 8000;

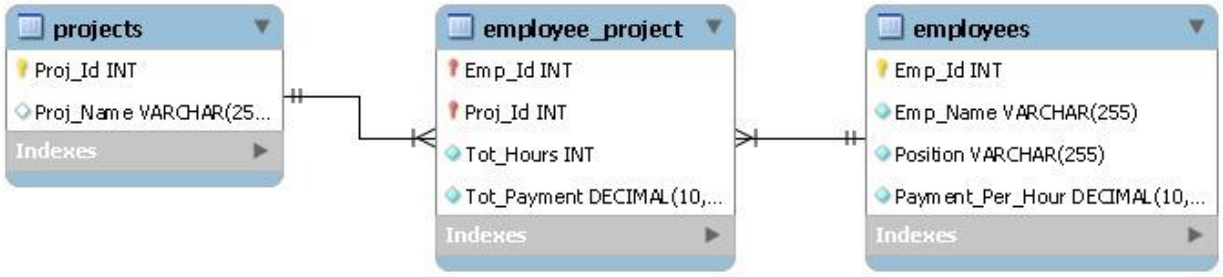
Here is how the table should look like:

Emp_Name	Tot_Payment
Bob	$3600
Tom	$3900
Bob	$5800

c) Let's Add 5% to the initial total payment for employees who worked more than 150 hours.

UPDATE Work **SET** Tot_Payment = Tot_Payment * 1.05 **WHERE** Tot_Hours > 150;

d) Let's build the ERD from the normalized database



2. Let's Identify Issues with the Table

1. **Data Redundancy**:
   - Course_Name, Lect_Name, and Lect_Email are repeated for every student in the same course.
   - Student information like St_Name is repeated for each course they enroll in.
2. **Multivalued Dependencies**:
   - Lect_Name and Lect_Email depend on Course_Id, not directly on St_Roll_No.
3. **Partial Dependencies**:
   - The current structure is not in **2NF** because Course_Name, Lect_Name, and Lect_Email are dependent only on Course_Id.

Students table:

St_Roll_No	St_Name
10	Peter
20	John
30	Dinah
40	Bob

Courses table:

Course_Id	Course_Name	Lect_Name	Lect_Email
1	Ml	Simon	sim@learn.cn
2	AI	Anna	an@learn.cn

Enrollments table:

Enrollment_Id	St_Roll_No	Course_Id	Grade
1	10	1	B
2	20	1	A
3	10	2	B
4	20	2	B
5	30	1	C
6	40	1	C
7	40	2	C

b) Let's build the ERD from the normalized database

